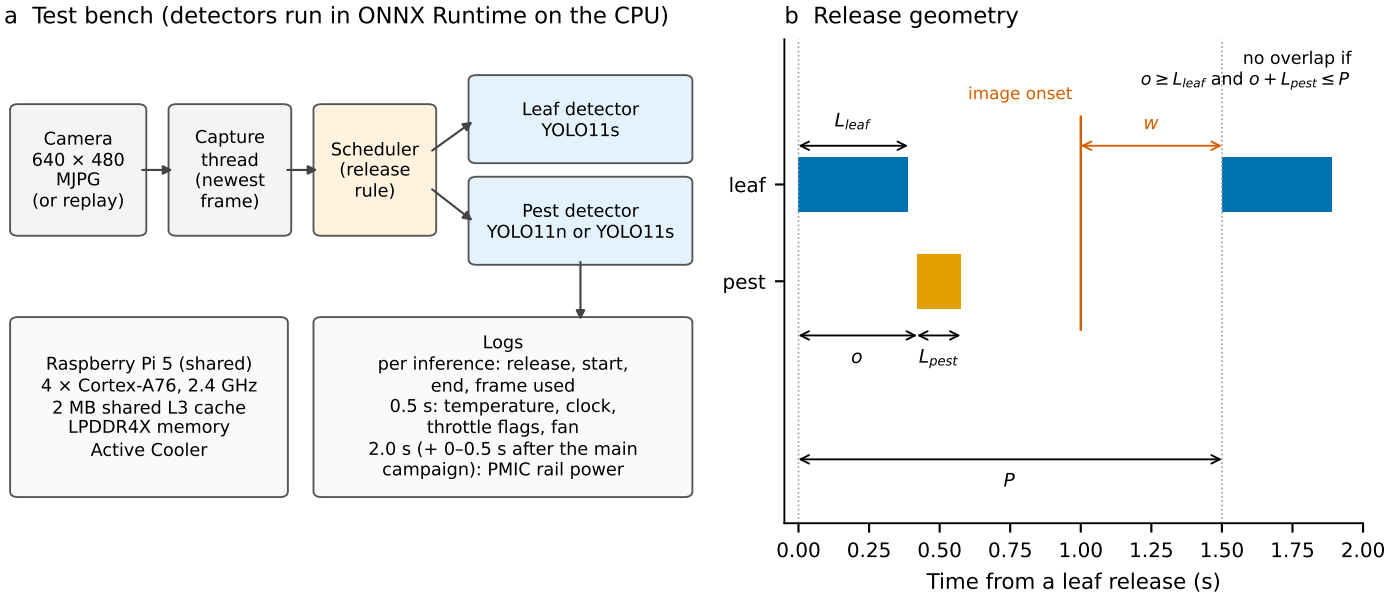
a Test bench (detectors run in ONNX Runtime on the CPU)
Camera
640 × 480
MJPG
(or replay)
Capture
thread
(newest
frame)
Scheduler
(release
rule)
Leaf detector
YOLO11s
Pest detector
YOLO11n or YOLO11s
Raspberry Pi 5 (shared)
4 × Cortex-A76, 2.4 GHz
2 MB shared L3 cache
LPDDR4X memory
Active Cooler
Logs
per inference: release, start,
end, frame used
0.5 s: temperature, clock,
throttle flags, fan
2.0 s (+ 0–0.5 s after the main
campaign): PMIC rail power
b Release geometry
no overlap if
$o \geq L_{leaf}$ and $o + L_{pest} \leq P$
image onset
L_{leaf}
w
leaf
pest
o
L_{pest}
P
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time from a leaf release (s)